



We and It: An interdisciplinary review of the experimental evidence on how humans interact with machines[☆]

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ABSTRACT

Today, humans interact with automation frequently and in a variety of settings ranging from private to professional. Their behavior in these interactions has attracted considerable research interest across several fields, with sometimes little exchange among them and seemingly inconsistent findings. In this article, we review 138 experimental studies on how people interact with automated agents, that can assume different roles. We synthesize the evidence, suggest ways to reconcile inconsistencies between studies and disciplines, and discuss organizational and societal implications. The reviewed studies show that people react to automated agents differently than they do to humans: In general, they behave more rationally, and seem less prone to emotional and social responses, though this may be mediated by the agents' design. Task context, performance expectations and the distribution of decision authority between humans and automated agents are all factors that systematically impact the willingness to accept automated agents in decision-making - that is, humans seem willing to (over-)rely on algorithmic support, yet averse to fully ceding their decision authority. The impact of these behavioral regularities for the deliberation of the benefits and risks of automation in organizations and society is discussed.

1. Introduction

Early in the morning, James drives to work on the route picked out for him by his navigation system. On the highway, his car's blind spot assistant helps him avoid a potentially dangerous situation. He happily listens to a playlist of new songs selected for him by an app. An alert on his phone warns him about unusual activity on his bank account, but the issue is swiftly resolved with the help of the bank's chat bot. After passing the automated barrier to enter his office, James plugs his portable computer into the docking station on his desk, and starts his day.

While James's morning may seem like nothing out of the ordinary, it serves to illustrate something rather extraordinary: how frequent and how routine interactions between humans and automated agents (i.e., agents of non-human nature powered by technology) have become. Had someone read this paragraph a few decades ago, they might have assumed that our hero was James Bond. In contrast, today, automated agents are everywhere. They collaborate with humans in the workplace and in some cases even occupy managerial positions. They engage in

sales negotiations with humans, and trade with them on the stock market. They are involved in deciding whether a human should receive a job or be released from jail. And, the recent move into a far more digital way of working due to the global pandemic has increased the involvement of automated agents in professional life and decision-making even further.

What happens when interactions between humans turn into interactions between humans and increasingly capable automated agents? How do humans react to the growing involvement of automated agents in decision-making? The ubiquity of automation in today's world calls for an interdisciplinary effort to answer these questions (Rahwan et al., 2019). At the same time, the fast-growing number of studies, the wide range of methodologies used and a number of (seemingly) inconsistent findings make it difficult for interested researchers or decision-makers to gain an understanding of the existing evidence. In this review, we attempt to change this by synthesizing the findings of 138 experimental studies from a wide range of disciplines into a number of generalizable findings which describe how humans behave when they interact with automated agents. We also point out a number of questions which the

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existing evidence does not yet allow to answer, because of a lack of data or because of inconsistent results, and provide concrete suggestions for future research.

As this review will show, a number of behavioral regularities emerge from the empirical evidence on human-machine interactions. In [Section 2](#), a series of experiments document that humans interact with automated agents in social ways, yet differently than they do with fellow humans: In general, they exhibit less of an emotional response to the actions of automated agents, and are less concerned - but not completely unconcerned - with social rules of conduct when interacting with automated agents. This reduced emotional and social response can improve outcomes in interactions that benefit from increased rationality - for instance, people have been found to be more likely to reveal sensitive health information to an automated system. On the other hand, this effect can be harmful where emotions or social rules are crucial drivers and motivations for beneficial outcomes - for example, interacting with automated agents has been shown to increase the likelihood of engaging in unethical or self-serving behavior. Certain design features of automated agents may affect the level of emotional and social response they trigger: For instance, some experiments show that features which cause automated agents to be perceived as more human-like intensify the emotional and social response to these agents. Task type and context seem to matter: While experiments document that humans are willing to engage with automated agents in contexts perceived as analytical or objective, they seem reluctant to do so in more social or moral contexts. In addition to these general results, we discuss insights from a number of studies where automated agents assume specific roles in interactions of particular relevance for the workplace.

We structure the review by the type of role the automated agents may play: in [Section 3](#), automated agents appear as relatively independent agents who either collaborate with humans as their peers, or manage them as their superiors. Though this use of automated agents is growing, experimental studies on this type of interaction remain rare. For example, the existing evidence establishes that having an automated teammate crucially affects worker (and hence team) performance, yet documents multiple channels for this effect which may lead to both an increase and decrease of performance. With regards to automated agents taking managerial roles, we discuss studies which show that humans seem willing to accept automated managers - and, indeed, that automated managers may even have an advantage over humans in contexts perceived as analytical. In [Section 4](#), we consider interactions where automated-agents are involved in decision-making situations. Here the existing evidence seems inconsistent at first glance: some studies show that humans are averse to delegating decision authority to automated agents, others find them to be appreciative of automated advice, and sometimes even overreliant on automated decision-making supports. We discuss why this could be the case - again, separating the evidence by the specific role the automated agents play, namely that of a delegate or of a decision aid, will help here - and propose a number of concrete testable hypotheses to reconcile these findings. In particular, we will argue that the acceptance of automated decision-making depends on the distribution of decision authority, the context of the decision and on performance expectations.

The studies covered in this review are chosen according to two criteria: first, we cover studies which investigate what changes when humans interact with automated agents, with the benchmark being human-human interactions. Choosing to focus on the presence of automation in an interaction, we use a broad definition of what constitutes an "automated agent" (AA), without imposing any constraints either on the underlying technology nor on its sophistication (indeed, computers/computer systems, machines, robots, algorithms, AI systems, intelligent agents and so on can all constitute automated agents). This allows us to draw on a large range of studies spanning from the early years of automation up until recent discussions of sophisticated machine learning algorithms. It goes without saying that the increased sophistication of automation and the growing familiarity of people with

technology over the last decades may affect the way in which people interact with it. Interestingly however, many of the behaviors reported in this paper appear to be triggered by the switch from purely human-human interactions to human-AA interactions, rather than by specific aspects of the underlying technology. Our second selection criterion is that we cover studies which use experimental methods to investigate human behavior in human-AA interactions, with the aim to discuss causal effects as well as underlying mechanisms. This focus on human behavior *in interactions* implies that we do not discuss research e.g. on algorithm design, user experience (UX) or interface design, nor on the stated attitudes towards automation in general, the macroeconomic consequences of automation or on the ethics of automation. We do touch upon some heavily debated topics such as explainable AI or algorithmic fairness in the discussion for additional perspectives on the topic.

Following our goal of identifying underlying behavioral regularities regardless of disciplinary borders, our review covers a wide range of fields, including economics, management, psychology, marketing and consumer research, sociology, judgement and decision making, human-computer interaction, neuroscience, computer science, information systems, medicine and even aeronautics. As a result of the differences across these fields, our sample includes studies which elicit stated preferences, revealed preferences or use physiological measurements, which use or do not use incentives, and which resort to various subject pools. The Appendix provides short notes on the methodologies of all cited experiments, observational studies and literature reviews.

To sum up our selection strategy, when faced with the trade-off between breadth or depth, we opted for the former. Regarding the methodology used to identify the relevant studies, due to the important differences in terminology and keywords across disciplines and technologies, our above-described choices implied that the search of digital databases using keywords soon proved to be unsuitable. Instead, we chose to primarily resort to ancestry searching (alternatively called backward reference searching). That is, after we identified a particular study as relevant, we considered if the studies it cites may be relevant for the review, too. If several relevant studies appeared in the same outlet, we more systematically reviewed other publications in that outlet resorting to journal hand searching. Finally, we consulted with other researchers working in the field (following [Whittemore & Knafl, 2005](#)). To guard against publication bias and to include the most recent findings we deliberately include publicly available working papers (following [Thompson & Pocock, 1991](#)). It is important to note that, given the ambitious goal of the review and the number of covered disciplines, the list of included papers is by no means exhaustive, but covers main themes and will hopefully provide a useful starting point to an interested reader.

The contribution of this review is threefold. First, we provide an extensive overview of behavioral research spanning from early automation research to discussions of modern AI systems, in a form that may prove useful to both interested scientists and decision-makers in the field. We synthesize the evidence from numerous studies into a number of recurrent findings. Second, we scrutinize a number of seemingly inconsistent findings, and suggest a structured way to reconcile them. We are able to address such inconsistencies due to the broad range of experiments covered (for other literature reviews that focus in more detail on only specific strands of literature see e.g. [Jussupow, Benbasat, & Heinzl \(2020\)](#) on algorithm aversion and appreciation, [Burton, Stein, & Jensen \(2020\)](#) on algorithm aversion, [Parasuraman & Manzey \(2010\)](#), [Alberdi, Strigini, Povyakalo, & Ayton \(2009\)](#) on automation bias and automation-induced complacency or [March \(2019\)](#) on the use of computer players in economic experiments). With this, we hope to provide concrete and testable hypotheses which might allow future research to investigate the currently open question of why different experiments document very different reactions to automated agents. Finally, we link the empirical findings to the ongoing societal discussion about the benefits and harms of automation and discuss how behavioral research can inform effective regulation of automated decision-making and

introduction of automation in the workplace. With all of this, we aim to promote the dissemination of insights and ideas across disciplines, and to help provide grounds for evidence-based decision-making.

Figure 1 provides an overview of the structure of the paper and shows the kinds of interactions considered in the review. Section 2 starts by discussing experiments investigating general patterns in interactions between humans and automated agents. In Sections 3 and 4, we consider specific types of interactions: Section 3 focuses on interactions in which automated agents collaborate with humans in the workplace, either as their peers and teammates or as their superiors. Section 4 discusses experiments which investigate how humans react to the involvement of automated agents in decision-making, where AA act either as delegates or as decision aids. At the end of each subsection, we offer short summaries of the key findings of the mentioned studies. Section 5 draws upon findings of the review to offer some thoughts on future avenues of research by linking the existing evidence to the ongoing political and societal discussion about the risks and benefits of automation. Finally, Section 6 concludes.

2. Social interactions with automated agents

2.1. The perception of automated agents as social interaction partners

To start our discussion, let us first establish whether interactions between humans and AA can appropriately be considered as social interactions. Researchers have long documented that humans tend to create narratives around events (Bruner, 1991), and for instance attribute agency to inanimate objects (Heider & Simmel, 1944). In an early series of lab experiments in the aptly termed *computers are social actors* (CASA) paradigm, psychologists documented that people apply social rules and expectations to computers (for an overview see Nass & Moon, 2000; Reeves & Nass, 1996). Participants in these experiments applied gender stereotypes (Nass, Green, & Moon, 1997) and ethnic stereotypes (Nass, Fogg, & Moon, 1996) to computers. They reacted to automated feedback (De Laere, Lundgren, & Howe, 1998), preferred computers

arbitrarily marked as teammates (Nass et al., 1996) and reciprocated helpful acts by computers (Fogg & Nass, 1997). Reactions of reciprocity were even adjusted to prevailing cultural norms (Katagiri, Nass, & Takeuchi, 2001). Yet, when asked directly, most participants in these experiments denied that they felt that computers had a personality, nor that they felt computers warranted social or polite treatment (Nass & Moon, 2000).

Later experiments provided further evidence for the social treatment of machines. Participants were again found to apply gender stereotypes (Tay, Park, Jung, Tan, & Wong, 2013; Eyssel & Hegel, 2012; Tay, Jung, & Park, 2014) and racial stereotypes (Bartneck et al., 2018) to automated agents. They were also found to use social cues like smiles and silence fillers when interacting with automated agents (Aharoni & Fridlund, 2007), and to react to automated flattery (Lee, 2010). Given sufficient behavioral realism, social reactions can be triggered by both avatars (digital representations of humans) and virtual agents (interactive computer programs) (Von der Puetten, Krämer, Gratch, & Kang, 2010). A recent experiment even showed that robots can trigger actions of social conformity at a similar rate as humans - but only until participants lose trust in faulty robots (Salomons, van der Linden, Strohkorb Sebo, & Scassellati, 2018).

Importantly, these findings do not imply that social interactions between humans and automated agents are equal to those between humans - as this review will discuss at length. Indeed, the difference in how humans interact with fellow humans or with automated agents are visible in neurophysiological studies, which document that different areas of the brain are activated in these different interactions (see e.g. Krach et al., 2008; Chaminade et al., 2012; Sanfey, Rilling, Aronson, Nystrom, & Cohen, 2003; McCabe, Houser, Ryan, Smith, & Trouard, 2001). The reason for this might lie in the way humans perceive their respective counterparts' agency. In social interactions people engage in the process of *mentalizing* to infer the mental state and capacities of their counterparts (Frith & Frith, 2006). This deliberation seems to occur less with automated agents: the area of the brain connected to mentalizing is activated less when humans interact with automated agents (McCabe

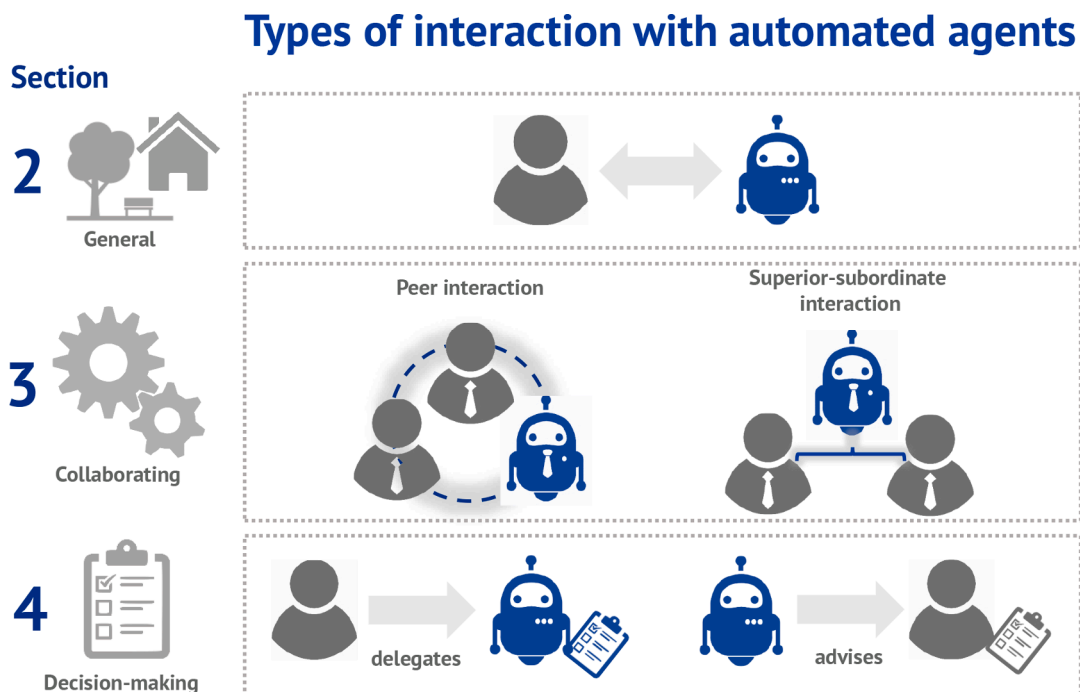


Fig. 1. Types of interaction between humans and automated agents: Illustration of different categories of interactions and the sections of the review which covers them. Within the collaboration section, interactions are separated according to the role of the automated agent as a peer (middle left) or superior (middle right). Within the decision-making section, the interactions are separated according to the distribution of agency within the interaction, with the human delegating the decision authority to the automated agent (bottom left), or the automated agent advising the human who retains the authority over the decision (bottom right).

et al., 2001; Coricelli & Nagel, 2009). The reduced need to infer the mental state of one's counterpart might also explain the repeated finding that people react faster to actions of automated counterparts (see e.g. Coricelli & Nagel, 2009; Farjam & Kirchkamp, 2018).

And, indeed, as a large survey shows, people do not attribute robots with minds (Gray, Gray, & Wegner, 2007). This survey in particular showed that humans deny robots the capacity to experience moral right or wrong (Gray et al., 2007). People even report feelings of unease when robots are described as able to feel (Gray & Wegner, 2012). However, despite this perception, people seem averse to mistreating automated agents. Indeed, people were found to exhibit both physiological and behavioral displays of stress when watching videos of robots being "tortured" (Rosenthal-von der Pütten, Krämer, Hoffmann, Sobieraj, & Eimler, 2013), or when asked to administer "painful" electric shocks to robots (Bartneck, Rosalia, Menges, & Deckers, 2005) or virtual agents (Slater et al., 2006) in two Milgram-style experiments (Milgram, 1963). Another experiment showed that people were unwilling to follow a command to destroy a tower a robot had built if the robot protested and started "sobbing" (Briggs & Scheutz, 2014).

Summary of section 2.1 Experimental studies have shown that people interact with automated agents in social ways, but differently than with other humans: When interacting with automated agents, people engage less in mentalizing (i.e. inferring the mental state of the counterpart). People do not seem to attribute robots with 'minds' or feelings, but are nevertheless found to be averse to mistreating automated agents.

2.2. The emotional and social response to automated agents

A robust finding which documents the differences between human-human and human-AA interactions is that interacting with automated agents triggers a reduced emotional and social response. This phenomenon has been documented using both subjective and behavioral measures, e.g. in trust games (Schniter, Shields, & Sznycer, 2020; Lim & Reeves, 2010) or in ultimatum, dictator and public goods games (Melo, Marsella, & Gratch, 2016). It has also been found using physiological measurements, e.g. when playing computer games (Mandryk, Inkpen, & Calvert, 2006), in ultimatum games (Sanfey, Rilling, Aronson, Nystrom, & Cohen, 2003; Van't Wout, Kahn, Sanfey, & Aleman, 2006) or in auctions (Teubner, Adam, & Riordan, 2015; Adam, Krämer, & Müller, 2015).

The reduced emotional response to automated agents is manifested both in terms of a less immediate emotional reaction to the agent's actions as well as in a generally decreased level of emotional arousal in human-computer interactions (as demonstrated e.g. in Teubner et al., 2015). Interacting with an automated counterpart seems to narrow the entire emotional spectrum: people react less positively to desirable actions by automated agents, and less negatively to undesirable ones. For instance, principals who delegated tasks reported significantly less enjoyment of good outcomes and significantly less anger for bad outcomes when the task was delegated to an algorithm rather than to a human (Leyer & Schneider, 2019). In another study, observers rated a physician's correct medical decisions less positively and incorrect decisions less negatively if the physician was described to use an automated decision-making aid (Pezzo & Pezzo, 2006). In a social exchange experiment, participants perceived coercive actions by computers as less unjust than coercive actions by humans, and retaliated against them less. In contrast, cooperative actions were perceived as similarly just regardless of who made them (Shank, 2012). The reduced emotional response could be due to a perceived lack of intent on part of the automated agent (as speculated by Shank, 2012). Hidalgo, Orghian, Canals, De Almeida, & Martin (2021) also highlight the role of intentions: in an investigation of almost 90 vignette scenarios with moral tasks, the authors find that identical decisions are judged differently depending on whether they are made by a human or an automated agents. The perception of intent seems to drive this difference: people

seem more likely to judge humans by their intentions and machines by their outcomes, resulting in more forgiving judgements towards automated agents in scenarios involving fairness, loyalty, and labor displacement.

A potential economic upside of this decreased emotional and social response is the increased rationality of interactions. The introduction of automated agents has, hence, been shown to reduce bubbles in simulated stock markets (Farjam & Kirchkamp, 2018), to increase bargaining efficiency in auctions (Adam, Teubner, & Gimpel, 2018) or to improve cooperation in ultimatum games (Erlei, Nekdem, Meub, Anand, & Gadiraju, 2020). Computers can also help avoid undesirable social responses, and, for example, increase the willingness to disclose uncomfortable information (Lucas, Gratch, King, & Morency, 2014). Reporting to computers has even been shown to significantly increase the likelihood of disclosures of intimate partner violence (Ahmad et al., 2009; Humphreys, Tsoh, Kohn, & Gerbert, 2011). On the downside, automation can prove harmful in contexts where beneficial behaviors are driven by emotions and social concerns like pro-sociality, social comparisons or reciprocity. An experiment which demonstrates this has participants play public goods, ultimatum and dictator games against other humans or against automated agents (Melo et al., 2016). In all three games, participants prove less willing to share with automated counterparts, and feel less guilty about exploiting them. This mirrors findings of earlier CASA studies, which documented participants' willingness to engage in self-serving bias with automated counterparts (i.e., to attribute positive outcomes to themselves and negative outcomes to the computer) (Moon & Nass, 1998; Moon, 2003). The lack of altruism and social pressure towards automated teammates was also shown to lower worker's productivity in an experimental sequential assembly line task (Corgnet, Hernán-Gonzalez, & Mateo, 2019). Further studies document that the presence of automated agents can increase the willingness to engage in unethical or even illegal behavior (for a review see Köbis, Bonnefon, & Rahwan, 2021). For instance, when given a chance to misreport the outcome of a coin toss to increase their monetary profit, participants were significantly more likely to lie when reporting to a computer (Cohn, Gesche, & Marechal, 2018). In summary, while automation can have beneficial effects in contexts where emotions or social concerns are detrimental, it can also be harmful by reducing them in contexts where they are beneficial.

The above cited studies that document a reduced emotional and social response aim to isolate what changes if the counterpart in an interaction is a human or an AA. These studies are generally conducted in a reduced setting. This in particular implies that, participants in these experiments are mostly not allowed to see or communicate with each other. Yet, other studies show that the social and emotional response to automated agents is mediated by, for instance, cultural background of a human counterpart (for an overview see Lim, Rooksby, & Cross, 2021) and by certain design features of automated agents that can render them to be perceived more human-like and emotionally intelligent. An important factor which affects a person's emotional and social response to an automated agent seems to be the agent's behavior and appearance. Indeed, humans seem to prefer interacting with automated agents which display contingent verbal and non-verbal reactions (Gratch, Wang, Gerten, Fast, & Duffy, 2007), relational and animated behaviors (Bickmore, Gruber, & Picard, 2005; Nitsch & Glassen, 2015) and group emotions (Correia, Mascarenhas, Prada, Melo, & Paiva, 2018). Further, Crandall et al. (2018) find that cheap-talk (costless, non-binding signals) allows to improve not only human-human cooperation but also to achieve a comparable level of human cooperation with a particular algorithm in a variety of cooperative games. Unexpected behaviors by robots can also amplify social responses: for example, people were more likely to display social reactions towards robots that unexpectedly cheated in a rock-paper-scissors game (Short, Hart, Vu, & Scassellati, 2010). In a study with service robots who delivered medicine to elderly patients, the use of human-like faces and voices were found to significantly promote positive emotional responses to the robots (Zhang et al., 2010). People

were also found to report feeling more comfortable around human-like robots and to perceive them as more useful (Castelo, 2019). The human-like appearance of an agent even affected the attribution of credit and blame in human-robot teams: automated teammates who appeared more human-like were relied upon more and attributed more credit for the output than those who didn't (Hinds, Roberts, & Jones, 2004). Cominelli et al. (2021) find that a humanoid appearance of the robot lead to an increased psychophysiological response as compared to the computer box. However, further studies show that the effects of human-like appearances remain unclear. An interesting counterpoint comes from an experiment which shows that anthropomorphic robots trigger more compensatory consumer responses - yet, not because people liked the human-like robots, but because they felt discomfort in their presence and perceived them as a threat to human identity (Mende, Scott, van Doorn, Grewal, & Shanks, 2019). And, as was mentioned earlier, participants in another study disliked robots who were described as able to feel (Gray & Wegner, 2012). More generally, a study on the roots of the behaviors documented by the CASA paradigm failed to find support for the claim that people are more likely to react socially to anthropomorphic characters (Lee, 2010). Nishio, Ogawa, Kanakogi, Itakura, & Ishiguro (2018) also find that it is not the appearance of the counterpart per se but rather the interaction of the anthropomorphic appearance and mentalizing stimuli that lead to reactions towards automated counterparts similar to human ones. Unfortunately, a full discussion of the effects of anthropomorphism is beyond the scope of this review. While the selected findings presented here aim to show that appearance seems to matter for social and emotional responses, we refer to other reviews for a full discussion of the factors involved (see e.g. Zlotowski, Proudfoot, Yogeewaran, & Bartneck, 2015), the ethical and societal impacts of anthropomorphic machines (see e.g. Darling, 2015) as well as challenges of affective computing (Picard, 2000).

Summary of section 2.2 In general, studies have shown that interactions with automated agents trigger a reduced emotional and social response in the humans involved. This reduction applies to both sides of the emotional spectrum, and may be beneficial, e.g. where it results in increased economic rationality or higher willingness to disclose sensitive information. It may also be harmful, e.g. where it leads to more anti-social or unethical behavior. The intensity of the emotional and social response depends on the characteristics of the human counterpart (e.g. their cultural background) and can be amplified by features of the AA (e.g. its appearance and behavior).

2.3. The importance of task type

Studies show that humans exhibit strong reactions when a previously human-human interaction is transformed into an interaction with an automated agent. An example of this comes from a field experiment which varied whether sales calls for financial services were made by humans or by chat-bots (Luo, Tong, Fang, & Qu, 2019). When the caller's identity was not revealed to the customer, chat-bots made about as many sales as experienced human sales workers. However, if the chat-bot's identity was revealed prior to the call, purchase rates fell by almost 80%. As the follow-up survey revealed, the customers claimed that the chat-bots were less knowledgeable and less empathetic - but only if they knew that they were interacting with chat-bots. However, both the identical sales rates and an analysis of the call data showed no performance differences. Similarly, Ishowo-Oloko et al. (2019) in a public goods game find that once participants are aware that their counterpart is a bot, they are less willing to cooperate - with the actual behavior of the counterpart having little impact.

Further studies point to a potential reason for this preference to interact with humans over automated agents: the type of task, and the perceived aptness of using automated agents for such tasks. Indeed, in an experiment investigating the attitudes toward the outsourcing of tasks to robots, people were found to react negatively to the outsourcing of social tasks to robots, but not to the outsourcing of tasks perceived as analytical

(Waytz & Norton, 2014). This distinction was confirmed by several other studies (see e.g. Lee, 2018; Hertz & Wiese, 2019; Castelo, 2019). Additionally, people tend to react to robots more positively if the appearance and demeanor of a robot match the task it is used for (Goetz, Kiesler, & Powers, 2003; Eyssel & Hegel, 2012). Such differences in the willingness to interact with automated agents depending on the type of task also appear in studies we will discuss later in the review (see in particular Section 3.2 and in particular Section 4.4).

Several studies considered human attitudes towards using algorithms in creative tasks. In an experiment featuring a poetry competition, judges were unable to reliably tell if a poem was composed by a human or an algorithm - yet rated the human poems higher and stated a stronger preference for human poems (Köbis & Mossink, 2021). Revealing the nature of the author did not affect the level of aversion exhibited in this study. Similarly, in Jago (2019), respondents rated creative works produced by algorithms as less morally authentic than the identical human works (note that this did not apply to what the authors call 'type authenticity', i.e. the accuracy of the work).

Summary of section 2.3 Studies have documented that people react negatively to automated agents in tasks regarded as social or 'human', but less so or not at all to their involvement in analytical tasks. The perceived aptness of automated agents for a task appears to drive these results.

3. Automated agents as peers and superiors in the workplace

The growing capacities and the increasing autonomy of automated agents (Haddadin & Croft, 2016) have vastly augmented the roles they can take in the workplace; from mere tools to be used by humans to increasingly autonomous collaborators or even managers in their own right. As this is a rather recent change, experimental research on how humans interact with such agents in the workplace remains rare. With the advancement of the technology frontier, these interactions are likely to grow in importance. In this section, we therefore discuss some interesting first studies which may help outline a future research agenda. In Section 3.1, we consider studies that look at the effects of automated agents who occupy the role of co-workers and teammates. In turn, Section 3.2 contains studies investigating the use of automated managers. Section 3.3 discusses open questions.

3.1. Automated agents as peers

The introduction of automated agents into existing organizations has so far mostly been investigated using qualitative methods. In particular, a number of observational studies show that the introduction of automated teammates goes beyond a simple replacement process and requires redefining roles, changing existing processes and implies a number of other challenges (see e.g. van den Broek, 2019; Lebovitz, Levina, & Lifshitz-Assaf, 2019; Stubbs, Hinds, & Wettergreen, 2007).

A first question here is whether humans are willing to use the help of automated teammates, i.e. to outsource some of their tasks to them. Inspired by the real-life experience of Walmart, a company which introduced robots to clean the shops, yet required workers to return to the cleaning tasks if the robots were unable to reach some areas, Alekseev (2020) investigated a setting where parts of a task can be automated, allowing the human workers to focus on the other parts. The study found that the share of people willing to use the automated mode was higher if the automated system could solve a higher share of the tasks. However, on average 30% of subjects chose not to use the automated system - even though it would have led to higher earnings. The main reasons workers gave for not using the automated agent were switching costs and a decrease in being able to focus on the task. This effect was dependent on the individual participants: players with lower cognitive abilities were more likely to opt out of using automation.

Performance Once a hybrid team is formed, a natural starting point for quantitative studies is to investigate the effect of automated agents

on the productivity and effort of their human co-workers and teammates. A few studies that consider this question conclusively find that there is a change in human performance, yet disagree on the direction of this effect. In the following discussion, we will attempt to speculate what may trigger the discrepancy in these findings.

In teams where team members execute a very similar task, having an automated teammate seems to decrease the performance of the human teammates (Corgnet et al., 2019; Dell'Acqua, Kogut, & Perkowski, 2020). This is documented, for instance, in a noteworthy lab experiment by Corgnet et al. (2019), where working alongside robots in a conveyor style production task reduces human effort. The authors establish that this reduction in performance is due to the reduced social response towards automated agents- namely, the lack of altruism and social pressure. Similarly, in Dell'Acqua et al. (2020), who consider hybrid teams in a coordination task, the authors demonstrate that even in a task where AI outperforms humans (and therefore should improve the team outcome), the performance of hybrid teams decreases in comparison to all-human teams. This decrease in performance is driven by an increase in coordination failures and a decrease in the effort provided by human teammates - though these negative effects are less pronounced for human players of high skill (similar to the findings of Alekseev, 2020). Thus, the effect of introducing automation on the performance of human teammates may depend on the type of workers in the team. In addition, the authors also report negative performance effects for all-human teams who worked alongside the hybrid team.

Yet, these negative performance effects might be mitigated by the human teammates' perception of who is responsible to complete a task. Two different studies consider a workplace interaction where the human and automated teammates perform somewhat different tasks, and find that automated teammates increase the effort humans put into the task. First, in a field experiment, Paravisini & Schoar (2013) find that credit officers extend more effort, i.e. spend more time deliberating a decision, when they expect an input (in the form of a score) from a decision-making support system. Similarly, when measuring task distribution in teams in the lab, Gombolay, Gutierrez, Clarke, Sturla, & Shah (2015) report that human teammates keep a larger share of the work to themselves when working with robotic teammates. A possible explanation for the increase in effort in Paravisini & Schoar (2013) and Gombolay et al. (2015) could be that the human subjects perceived themselves to be more responsible for the outcome, felt that they could not fully rely on the automatic teammate and hence increased their own efforts.

Responsibility A closely related question is what effect automated teammates have on the distribution of responsibility within a team. While the diffusion of responsibility in teams is a well documented phenomenon for human groups (see e.g., Darley & Latané, 1968), and one could hence expect that human players might readily shift responsibility to automated agents, in particular given the reduced social concerns against them, this does not seem to be the case. In a laboratory experiment using a modified dictator game, Kirchkamp & Strobel (2019) found no significant differences in taking responsibility between human-human and human-computer teams - though participants in human-computer teams did behave slightly more selfishly. That is, in these types of interactions the decreased social response seems to be balanced out by the lack of agency of automated agents. These results are in line with previous findings by Kim & Hinds (2006), who highlight that the level of autonomy of the robot is important to how much credit or blame is attributed to it. In a recent online experiment, Hohenstein & Jung (2020) considered if AI can be blamed for unsuccessful coordination between two human counterparts by having subjects chat with or without AI assistance (in the form of smart replies by the messenger program). If the team managed to coordinate, the human participants perceived the responsibility for success to be on them, but if it failed, some of the responsibility (ca. 10%) was attributed to the AI system. Interestingly however, participants did not use AI to deflect blame from themselves but rather to reduce the share of blame that they assigned to

the partner.

By refraining from shifting (more) responsibility towards computers, the participants forewent a seemingly effective way to avoid being held responsible themselves, as an earlier study showed: Observers of accidents tended to attribute less responsibility to a company if technology was involved in the accident (Kurtzberg & Naquin, 2004). Similarly, in another study observers attributed less fault to a doctor who followed the (bad) recommendation of an automated decision aid (Pezzo & Pezzo, 2006). Similar results can be found in a recent study by Liu & Du (2021). Two recent studies focus particularly on responsibility attribution in moral tasks; Feier, Gogoll, & Uhl (2021) focus on responsibility attribution in moral tasks, and find that in case of an unfavourable outcome, observers assign less blame to the delegator of a moral task if the task was delegated to a machine (rather than to another human). Delegators in the study are not aware that delegating to the algorithms is an effective way to shift blame. Shank, DeSanti, & Maninger (2019) considers the perception of moral violations produced by human-algorithm pairs, in two versions where a human either monitors an AI system or receives a recommendation by it. The authors find that the humans are held responsible for the outcome despite the involvement of AI, yet that humans who monitor AI are faulted less than humans who handle the task alone and than humans receiving AI recommendations. Further studies on this seem necessary, in particular to investigate why observers seem more ready to shift responsibility to automated teammates than to human teammates, and why teammates themselves do not seem to use automated agents as scapegoats even though this seems like an efficient way to deflect responsibility.

Social Cohesion Finally, experiments show that having an automated teammate may affect the interaction among the humans in the team (Traeger et al., 2020; Short & Mataric, 2017; Strohkorb et al., 2016). For example, one study found that the human teammates were more likely to engage in social conversations and perceived the group more positively if the robotic teammates expressed vulnerability (e.g. by admitting mistakes) (Traeger et al., 2020). Robots might thus be able to help people collaborate better. However, these studies did not compare the reactions triggered by the robots to those that could be triggered by similarly acting humans in the group. As Rahwan, Crandall, & Bonnefon (2020) point out, the strategies that were successfully employed by automated agents in this study are unlikely to work for a human. Based on the finding that automated agents elicit less social responses than humans (see Section 2.2), it might be the case that the social cohesion triggered by robots is lower than the one that would be triggered by a human. Yet, it is indeed notable that these studies show that automated agents can act as social catalysts; a finding which may inspire another dimension of studies in human-AA interaction.

Summary of section 3.1 The experimental evidence shows that the introduction of automated teammates can affect the performance of human teammates both negatively and positively. Performance decreases may be due to reduced social reactions to automated agents and/or to coordination difficulties. Performance increases may be due to higher perceived responsibility of the human for a task. Differences in the cognitive abilities of the human partners are important for the willingness to engage and the success of interacting with automated agents. Teammates in hybrid teams do not appear to shift blame to their automated counterparts. However, third party observers seem to fault humans less if an automated agent is involved. The autonomy of the automated agent matters for how blame is attributed. Automated agents can affect how human team-mates interact among each other within the team.

3.2. Automated agents as superiors

Today, an increasing number of managerial tasks are performed by automated agents (see e.g. Aharoni & Fridlund, 2007; Lee, 2018). Unsurprisingly, this has started to receive attention in behavioral research, and there is some first experimental evidence on whether and when

humans are willing to accept automated managerial decisions.

First, experiments have documented instances where people are willing to accept automated managers. Cormier, Newman, Nakane, Young, & Durocher (2013) demonstrated that people are willing to obey automated managers: In an obedience experiment, the manager instructing participants to complete a tedious task is either a human or a robot. While participants protested significantly more frequently, earlier and quit the task sooner when instructed by a robot, they were still willing to obey it. Indeed, about half of the participants in this condition continued with the tedious task for the full duration of the study, even after trying to protest. In a field experiment with Alibaba warehouse workers, a substantial share of participants were even found to prefer automated managers to human ones (Bai, Dai, Zhang, Zhang, & Hu, 2020). Here, warehouse workers received lists of items to collect from shelves either from a human manager or from a computer terminal. The workers in the experiment generally perceived the lists distributed by an algorithm through the computer terminal to be fairer than those handed out by human managers (although both were generated by the same algorithm). As a result of differences in perceived fairness of task assignment, warehouse productivity increased in the automated conditions. A lab experiment using a lego set assembly task further documented that people are content to have a robot allocating tasks within the team, and may even assign them such managerial responsibilities themselves (Gombolay et al., 2015). Additionally, this study finds that both subjective and objective measures of participants' satisfaction increased with increased autonomy of the automated agent. Workers preferred to have the robot make scheduling decisions, and spent less time on rescheduling tasks the more autonomous the robot was.

However, note that another experiment in a similar setting finds the opposite result, namely that participants disliked robotic managers, were more likely to blame them for mistakes, and relied upon them less (Hinds et al., 2004). Importantly, while in Gombolay et al. (2015) the robots actually performed managerial tasks (e.g. scheduling), in Hinds et al. (2004) they performed supporting tasks within the team (e.g., carrying the parts), and were exogenously assigned managerial status. Hence, as speculated by the authors in Hinds et al. (2004), the aversion to robot managers may be due to a mismatch in skills and authority, not due to the robotic nature of the supervisor itself.

The acceptance of automated managerial decisions might also depend on the context of the task. Accordingly, Lee (2018) finds that human and automated managerial decisions are perceived as equally fair for analytical tasks - but not for social tasks, where the human decisions are perceived as fairer. Similarly, Newman, Fast, & Harmon (2020) also document that the use of AI for more social managerial tasks might not be very well accepted. In a series of vignette experiments, the authors consider how employees perceive a variety of HR decisions (layoffs and promotions, performance review, determining the size of the bonus, evaluating interviews) if they are made by an algorithm. In contrast to the findings of Bai et al. (2020), their results document that people perceive algorithmic decisions as less fair, consider them to be reductionistic, leading them to think that certain qualitative information or contextualization is not being taken into account.

Finally, with regards to automated performance evaluation and bonus assignment, Strobel (2019) finds that employees exert significantly more effort if the threshold performance for receiving a bonus is set by a human manager manually on case by case basis as compared to setting the threshold *ex ante*, before performance of individual employees is known and passing on to an automatic system. However, this result appears not to be driven by fairness considerations or task context, but by the fact that employees expect a lower performance threshold in the automated condition (i.e., expect to receive a bonus even with lower effort).

Summary of section 3.2 The experimental evidence shows that humans can accept automated agents performing managerial tasks, and that in some contexts automated managers may even be preferred over human ones. An important factor driving the reactions to automated

managers is the type of the task (analytical vs. social).

3.3. Open questions

In summary, the cited experiments on human-machine interaction in the workplace show that people behave differently in the presence of automated co-workers or managers, and that there may even be spillovers into the interactions between humans on the team. It seems that there are several behavioral channels through which the introduction of an automated teammate may negatively affect the performance of human co-workers. Yet, the perception of responsibility for the task appears to be a channel that might mitigate or even override the negative effect. As for automated managers, it appears that the nature of the managerial task plays a particularly crucial role in the reactions to automated managers.

Given that managerial tasks carry special importance by affecting organisational success at large as well as productivity of individual employees, it seems to us that a particularly promising avenue of research is to investigate where automated or human managers might be particularly suited. Based on the findings discussed in Section 2 and the evidence above, automated managers may be especially suitable for more analytical tasks and have an edge over human ones when it comes to making unpopular decisions, as these might trigger less of an emotional response in the human subordinates. On the other hand, the decreased social response to both automated managers and teammates might carry the problem of increased misreporting and unethical behavior. This concern may be counterbalanced by the appearance and anthropomorphism of the agents. However, this in turn may affect the perceptions of agency and responsibility within the team.

In conclusion, more systematic research is needed in order to understand how we can leverage the peculiarities of human-machine interaction in the workplace. As technology advances rapidly, developing our body of knowledge on the topic of hybrid interaction in the workplace and group dynamics is of high urgency.

4. Automated agents as delegates and aids in decision-making

A topic that has received particular attention in the literature is the use of automated agents in decision-making. The experiments discussed in the following section all contribute to the understanding of whether humans are willing to accept the involvement of automated agents in decision-making - and, if so, to what extent? Are they only willing to accept automated decision aid, or are they willing to fully delegate decisions to them? Yet, although the research questions of the papers in this section are similar, the results often are not. Indeed, as we will discuss in this section, the experiments will document different behavioral patterns, and in consequence, will sometimes come to different conclusions with regards to how humans perceive the use of automation in decision-making: In particular, several studies document that humans are averse to delegating decision tasks to automated agents, a phenomenon called *algorithm aversion*. Other experiments find that humans prefer automated advice to human advice, a phenomenon called *algorithm appreciation*. Yet further studies find that humans over-rely on automated decision-making supports, failing to correct for their mistakes (*automation bias*) or to properly monitor them (*automation-induced complacency*).

At first glance, these results seem puzzling - how can people simultaneously be averse to automated decision-makers, and prefer their advice to human advice? However, as we will discuss in the following sections, there are a number of factors which might explain these patterns in a consistent way. In an attempt to reconcile these seemingly inconsistent findings, we categorize the contexts in which they occur according to the distribution of agency within the interaction. That is, we propose to distinguish between situations in which the primary or exclusive authority over the decision is delegated to the automated agent (Section 4.1), and those in which the human retains the primary/

exclusive authority over the decision (Sections 4.2 and 4.3). As we will discuss, the experimental findings suggest that this distinction in agency is of major importance for how humans react to algorithmic decision supports. In the final Section 4.4, we will also discuss further potential factors which might explain the inconsistencies in findings.

4.1. The aversion to automated agents as delegates

Let us first consider a situation which is currently steadily increasing in importance: Decision-making where the authority over the decision is ceded to an automated agent. This cession of authority can be done fully - think, for example, of an automated trading program, which is set up initially by the user, but then makes autonomous sales and acquisitions on the stock market. It can also be done only partially - think, for example, of an algorithm choosing a small selection of candidates to be interviewed by a human employer. In both of these examples, an important part of the decision is delegated to the automated agent. The question which arises here is whether, and under what circumstances, humans are willing to make this delegation.

In an influential series of experiments investigating this question, participants were tasked with a number of forecasting tasks (regarding the future success of MBA graduates and the future development of airline passenger counts), and given the choice between making these predictions themselves or delegating the task to an algorithmic forecaster (Dietvorst, Simmons, & Massey, 2015). The participants were aware that the algorithm had made some mistakes in the forecasts - but it still consistently outperformed the human participants. Nevertheless, the participants were more likely to choose to make the prediction themselves. This costly choice is an example of a behavior the authors of the study termed *algorithm aversion*. The preference of humans to rely on themselves rather than to use superior automated decision-making supports had also been documented previously in other contexts, including when driving a car (Kantowitz, Hanowski, & Kantowitz, 1997) or when solving a visual detection task (Dzindolet, Pierce, Beck, & Dawe, 2002).

Importantly, algorithm aversion does not simply imply that people prefer to make decisions themselves. Given the choice between receiving advice from a human or an automated agent, people were more likely to go with the human for joke recommendations (Yeomans, Shah, Mullainathan, & Kleinberg, 2019), for medical tasks (Promberger & Baron, 2006; Longoni, Bonezzi, & Morewedge, 2019) and when the task was unknown (Hertz & Wiese, 2019). Doctors were perceived as less qualified by observers if they delegated decisions to automated decision support, but not when they sought advice from human colleagues (Arkes, Shaffer, & Medow, 2007; Shaffer, Probst, Merkle, Arkes, & Medow, 2013).

Algorithm aversion may affect decision-making: In experiments, participants were more likely to follow medical advice when it came from a human provider rather than from an AI agent (Longoni et al., 2019), and gave more weight to investment advice by humans rather than a statistical forecast (Önkal, Goodwin, Thomson, Gönül, & Pollock, 2009). In an observational study, managers at a clothing retailer were prone to reject a decision support system's advice regarding markdowns for sales (Saez de Tejada Cuenca, 2019). And, indeed, the debate about when people would do better if they followed algorithmic advice dates back decades (Meehl, 1957; 1954).

Multiple possible causes of algorithm aversion, or more generally of the "disuse, or the neglect or underutilization of automation" (Parasuraman & Riley, 1997), have been proposed by the literature. A first candidate is low trust in automated agents. Indeed, people were found to be particularly unlikely to use automated decision-making aids in uncertain domains (Dietvorst & Bharti, 2019), where trust is arguably more important. In the study on the use of automated investment advice, participants stated that they trusted automated agents (slightly) less - though this did not predict whether they actually followed the automated advice (Önkal et al., 2009). A potential reason for the lack of trust

in automation is advanced by B. Dietvorst and coauthors (Dietvorst et al., 2015; 2018; Dietvorst & Bharti, 2019), who argue that the cause of algorithm aversion is seeing an algorithm err. According to this argument, while people are willing to forgive humans for making mistakes, this leniency is not extended to algorithms. Indeed, earlier studies also show that while participants initially trusted automated agents more than humans and preferred to use them, they lose trust quickly and prefer human aids after seeing the automated agents make mistakes (Dzindolet et al., 2002). Machine errors have been shown to cause low trust in their performance, particularly if error rates are not constant (Muir & Moray, 1996). Particularly strong support for the argument of the loss of trust after seeing an algorithm err comes from a more recent study, which finds no support for the hypothesis that human advice is generally preferred to automated advice, but shows that after receiving incorrect advice the utilization of automated advice decreases significantly more than the utilization of human advice (Prah & Van Swol, 2017).

However, findings from other studies cast doubt on these arguments. For instance, the resistance to the use of an AI health provider documented in a series of experiments persisted even when the AI agent was specifically described as being superior to the human regarding the number of complications/accurate diagnoses (Longoni et al., 2019, study 3c). Neither the (insignificant) differences in trust extended towards automated agents or humans nor the perceived utility of the agent were able to explain the preference for the human decision-maker documented in another experiment (Gogoll & Uhl, 2018). And, two studies of social robots even found that people did either not mind the robots' mistakes (Salem, Lakatos, Amirabdollahian, & Dautenhahn, 2015), or even preferred robots who made mistakes to those who performed flawlessly (Mirnig et al., 2017). Note that, presumably, these last findings were related to the task context: While the above mentioned studies which document salience of machine errors mostly used analytical tasks (e.g., Dietvorst et al., 2015; Önkal et al., 2009), these robots were used for social tasks. It seems possible that there is some kind of interaction effect between task characteristics and sensitivity to errors: Humans may not be willing to accept automated mistakes in analytical contexts, but they seem to accept or even prefer fallible machines in more social contexts - after all, it would only make the machines appear more human. More generally, the findings of automation bias and algorithm appreciation to be discussed in Sections 4.3 & 4.2 show that people do not by and large distrust machines. Indeed, as we will discuss, these papers document instances where humans trust automated agents too much. Trust might also depend on the anthropomorphic appearance of the automated agent (for an example with autonomous vehicles see Waytz, Heafner, & Epley, 2014, for a general overview of the topic of trust in machines see e.g. Lee & See, 2004; Glikson & Woolley, 2020).

A second potential cause of algorithm aversion, which is advanced by another experiment (Gogoll & Uhl, 2018), is the moral context of the decision. The authors define a moral decision as a decision which affects a third party. In this experiment, participants are given the choice between delegating a calculation task to another human or to an algorithm. The earnings for the task are given to a third person, placing the task in the moral domain. Participants were not only significantly more likely to go with the human decision-maker, but even punished others who chose the algorithm. As neither the trust extended towards the agent (measured using a trust game), nor its perceived utility can explain these findings, the authors argue that people might exhibit a *per se* aversion to the use of automated decision-making agents in the moral domain (as stated, defined as decisions which may affect third parties, e.g. decisions by automated driving systems). This is corroborated by another series of experiments which also documents that humans are averse to automated agents making moral decisions ("morally-relevant driving, legal, medical, and military decisions", Bigman & Gray, 2018, p. 21). Interestingly, this aversion holds irrespective of whether the decisions made are favorable for the third party affected by them. A

potential insight into the reasons behind this preference comes from Longoni et al. (2019): Here, the authors propose that the cause of algorithm aversion is “uniqueness neglect”, i.e. the concern that an automated agent is unable to account for an individual’s unique characteristics and circumstances. Testing this hypothesis in the healthcare context, the authors find that a participant’s perceived sense of uniqueness is able to predict their aversion to the use of medical AI. Hence, these studies show that the moral context of a decision might be the factor driving algorithm aversion. However, this explanation does not seem satisfactory for the aversion documented in the studies by B. Dietvorst and coauthors: Here, algorithms are used to forecast the success of MBA graduates and the number of airline passengers (Dietvorst et al., 2015), standardized test scores (Dietvorst, Simmons, & Massey, 2018) and for a number of deliberately abstract judgement tasks (Dietvorst & Bharti, 2019) - all tasks outside of the moral domain.

However, we feel that there is another important factor to note. In almost all the studies we cited above, the participants are found to exhibit algorithm aversion when they are given the choice between a human or an automated decision-maker (Dietvorst et al., 2015; Gogoll & Uhl, 2018; Promberger & Baron, 2006; Yeomans et al., 2019; Hertz & Wiese, 2019; Önköl et al., 2009, is a notable exception). In other words, algorithm aversion seems to occur where humans cede the authority to make a decision to an automated agent. However, a different situation arises when an automated decision-making system is used not to replace the human decision-maker, but to assist him. As we will discuss in the following two sections, there is evidence which suggests that this factor may not only remedy algorithm aversion, but even trigger a positive preference of automated over human decision-making support.

Summary of section 4.1 Experimental studies show that when able to delegate a decision task to an automated agent, people frequently exhibit *algorithm aversion*, i.e. prefer to make the decision themselves or to delegate it to another human. Some studies link this aversion to lower trust and lower performance expectations in AA compared to humans. However, algorithm aversion has also been documented in cases where the AA has and is perceived to have superior performance and where there seems to be no difference in trust extended. Algorithm aversion seems to be particularly pronounced in moral contexts, and might be caused by the perception that machines are unable to account for human individuality. As we will discuss in the next section, if AA is in a supporting role to a human, some studies find the opposite, i.e. document instances where automated decision support is preferred over human support.

4.2. The over-reliance on automated decision aids

We now consider a different kind of the use of automated agents in decision-making: to assist and support, but not to replace the human decision-maker. Examples of such settings could be a pilot flying a plane with the help of an autopilot, the operator of a machine being assisted by an automated alert system, or a person drafting a text being supported by a spell-checking software. A series of experiments, mostly from psychology or applied engineering fields, have documented that people frequently over-rely on the automated agent’s advice in these settings, fail to realize when the automated advice is wrong and fail to intervene when they should (for reviews of this literature see e.g. Parasuraman & Manzey, 2010; Alberdi et al., 2009). This phenomenon is termed *automation bias*, or the over-reliance on automated decision-making support. Note that, in fact, this literature describes two related phenomena: *automation bias*, i.e. the failure to intervene, and *automation-induced complacency*, i.e. the failure to appropriately monitor automated support. We will refer to both of these phenomena using the broader term of *automation bias* (following Parasuraman & Manzey, 2010).

Automation bias occurs when people treat the automated agent “as a heuristic replacement for vigilant information seeking and processing” (Mosier & Skitka, 1996, p.205). It manifests itself in errors of commission, i.e. misguided actions based on a false alarm by the automated

system, or errors of omission, i.e. the missing of critical events when the automated system fails to flag them (Skitka, Mosier, & Burdick, 1999). In one of the earliest studies of this phenomenon, pilots were found to be more easily misled by (faulty) automated checklists than by simple paper checklists (Mosier, Palmer, & Degani, 1992). This result was then replicated with a flight simulation task in the lab (Skitka et al., 1999). Here, participants in automated conditions were more likely to miss events not flagged by the software (i.e. to commit errors of omission), and more likely to act on incorrect software advice (errors of commission) compared to flying under manual control. Both laypeople and experts are susceptible to automation bias, as was documented in a number of flight simulation studies with experienced pilots (Mosier, Skitka, Heers, & Burdick, 1998; Sarter & Schroeder, 2001; Galster, Duley, Masalonis, & Parasuraman, 2001). Similarly, in a study with experienced air traffic controllers, significantly fewer controllers detected a conflict that the system had failed to flag compared to when conflicts were handled manually (Metzger & Parasuraman, 2005).

Later studies document the over-reliance on automated decision support outside of the domain of aviation. For instance, the use of a spellcheck program was shown to prompt people with high writing skills to miss spelling errors and to falsely change correct spellings (Galletta, Durcikova, Everard, & Jones, 2005). In health care, the reliance on incorrect automated advice was shown to lower the decision accuracy of physicians reading EKG charts (Tsai, Fridsma, & Gatti, 2003). The sensitivity of professional film readers reading mammograms decreased if they used computer-aided detection systems - causing them to miss cancers if the system had failed to mark them (Alberdi, Povyakalo, Strigini, & Ayton, 2004). This result was replicated in a follow-on study which in addition found that non-cancerous objects were more likely to be marked as cancers if the system had falsely flagged them as such (Alberdi, Povyakalo, Strigini, Ayton, & Given-Wilson, 2008). Further, automation bias and automation-induced complacency have also been documented in process control (Bahner, Elepfandt, & Manzey, 2008; Manzey, Reichenbach, & Onnasch, 2012; Wickens, Clegg, Vieane, & Sebok, 2015) and command and control situations (Rovira, McGarry, & Parasuraman, 2007). In summary, the evidence on automation bias shows that while automated decision-making aids can improve decision quality when they are correct, they may also lower decision quality when they are incorrect, because people are nevertheless prone to rely on them.

Automation bias seems to be more likely to occur in situations of high cognitive load, which may stem from task complexity, multitasking or from time pressure (Parasuraman, Molloy, & Singh, 1993). Notably, experience with automated systems or with sharing the load with teammates do not seem to remedy automation bias (Skitka, Mosier, Burdick, & Rosenblatt, 2000b; Mosier, Skitka, Dunbar, & McDonnell, 2001). Indeed, automation bias seems not to stem from general inattention, but from the monitoring of the automated agent having lower priority and receiving less attention than other competing tasks (Parasuraman & Manzey, 2010). In line with this, an experiment has documented that providing people with variable priority attention training reduces automation bias (Metzger, Duley, Abbas, & Parasuraman, 2000). A further effective way to reduce automation bias seems to be highlighting the responsibility and accountability of the monitoring person (Goddard, Roudsari, & Wyatt, 2012; Skitka, Mosier, & Burdick, 2000a).

Summary of section 4.2 Experiments have documented that when people are supported by an automated agent in a decision task, they are likely to exhibit *automation bias*, i.e. to over-rely on the automated support and to (fail to) intervene following its incorrect inputs. Automation bias may occur irrespective of the level of experience of the user. Studies show it to be more likely in situations of high cognitive load. Automation bias can be mitigated by prompting higher priority to monitoring the system by i.e., highlighting the responsibility and accountability of the human, or by training attention capabilities.

4.3. The preference for automated decision aids

The decision-making situation in which a human is assisted by an automated agent has also started to receive attention in the behavioral economic literature. However, the focus of these studies is slightly different to the ones investigating automation bias: here, the question asked is not how a human will behave when assisted by an automated agent, but whether humans want such automated assistants. For example, think of a physician who can choose to use an automated diagnostic support system to help diagnose patients, or of a person asked to make a prediction and given the option of receiving automated support for this. That is, in contrast to the settings of the experiment documenting algorithm aversion discussed in Section 4.1 and as in the experiments discussed in Section 4.2, here the automated agent does not replace the human decision-maker, but assists them. The few economic studies which investigate this situation have shown that, in this set-up, humans are appreciative of automated advice and may even prefer it to human advice - a phenomenon termed *algorithm appreciation*.

In the leading series of experiments on algorithm appreciation, Logg, Minson, & Moore (2019), who also coined the term, ask participants to make a number of quantitative judgments (visual estimation of a person's weight, prediction of physical attractiveness, prediction of song popularity) under uncertainty, and are offered advice from either humans or automated agents (Logg et al., 2019). The participants consistently chose to receive and put more weight on automated advice, rather than human advice. Hence, the authors conclude, people seem appreciative of automated decision-making support. Further empirical support for this appreciation of automated advice is found in an earlier experiment by Dijkstra (1999): Here, participants are provided with (correct) human advice and (incorrect) algorithmic advice for a legal case. The participants very frequently relied on the algorithmic advice, in particular when it was given in production rule form (the latter replicating a previous finding from Dijkstra, Liebrand, & Timminga, 1998). Outside of the lab, people were found to be responsive to algorithmic advice when choosing health care plans in a randomized control trial (however, this study does not feature a comparison to the responsiveness to human advice, Bundorf, Polyakova, & Tai-Seale, 2019).

Interestingly, a close look at the leading experiment on algorithm aversion by Dietvorst et al. (2015) shows that participants in the control condition, who did not see the model perform, also stated that they were more confident in the model's, rather than in the human, forecasts. More generally, the finding that humans appreciate algorithmic advice is supported by a number of results contained in studies which generally document algorithm aversion, but find that this aversion is decreased or even eliminated when the agent's role is framed as only advisory. Indeed, for example, in Dietvorst et al. (2018), people are more likely to use a forecasting algorithm when they are able to intervene and even just slightly modify its advice. Other studies show that participants are less averse to the use of automated agents in medical and other moral decisions when the agent provides the humans with a recommendation, rather than a decision (Promberger & Baron, 2006; Bigman & Gray, 2018). Similarly, a vignette study in a consumer and medical setting finds that participants appreciate professionals' use of automated decision aids, but react negatively to a full delegation of decision authority (Palmeira & Spassova, 2015). Finally, framing a medical AI system as providing support rather than replacing the human health care provider fully eliminates the algorithm aversion documented in the rest of a series of experiments in (Longoni et al., 2019, study 9).

Summary of section 4.3 Experimental studies have shown that, when people are assisted in making a decision by an automated agent, they can exhibit *algorithm appreciation*, i.e. be appreciative of automated advice and may even prefer it to human advice. The shift from algorithm aversion (see Section 4.1) to algorithm appreciation seems to occur when people are allowed to intervene and adjust the recommendation of the automated agent.

4.4. Open questions

In summary, while the studies discussed in Section 4.1 document instances where humans are averse to the use of decision-making algorithms, the studies discussed in Sections 4.2 and 4.3 document instances where humans are indifferent between human or automated decision-makers, or even prefer and/or over-rely on automated rather than human decision advice. What emerges is that there is a clear need for future, integrative studies which investigate the factors triggering one or the other phenomenon. In the following we deliberate on a few hypotheses that such studies could investigate.

First, as we have already alluded to, a factor which could trigger algorithm aversion in one case, and algorithm appreciation or automation bias in the other, is the distribution of agency in the interaction. There is evidence that people seem particularly appreciative of automated agents when they are framed as providing support rather than as independently making the decision. To cite but one example, in one of the vignettes of a study that generally documents the aversion of people to machines making moral decisions, participants are asked to choose who they would prefer to make a moral decision: a human doctor, an "autonomous statistics-based computer system", or the doctor advised by the automated system (Bigman & Gray, 2018, p. 26). The participants were most likely to choose the last option. In the words of the authors, "[t]hese results suggest that most people are willing to have machines involved in moral decisions, as long as they are not the ones to make the actual decisions" (Bigman & Gray, 2018, p. 30). More generally, the empirical evidence seems to suggest that humans are averse to fully giving up their decision authority, yet appreciative of automated advice when they retain - or when they feel that they retain - the ultimate authority over the decision. However, this hypothesis so far rests on a small number of studies and some corollary results, while studies which purposefully investigate the impact of this factor remain lacking. Hence, it remains unclear whether the exact degree of human involvement matters - does the human need to retain real determinative capacity, or is it sufficient if humans feel like they remain involved? And, if such nominative human involvement were sufficient to avoid algorithm aversion, would it be ethically and/or legally defensible to mislead people into thinking that they remain involved in decision-making?

A second factor that might explain the discrepancy is the context of the decision, respectively the type of task contained in the decision. In particular, the empirical evidence we have discussed suggests that while humans are willing to accept automated agents in areas considered more objective or analytical, they are more reluctant to do so in areas considered as social or moral. This could be due to normative preferences, i.e. the view that some decisions simply should be made by humans (see e.g. Gogoll & Uhl, 2018, who speak of a *per se* aversion to the use of algorithms in the moral domain). On the other hand, it could also be due to performance expectations, i.e. the view that humans are simply better at making social or moral decisions. A formulation of this argument comes from the study by Longoni et al. (2019), who argue that 'uniqueness neglect', the view that automated agents are unable to account for the uniqueness of humans, is the cause of algorithm aversion. In other words, people might be averse to the use of decision-making algorithms in social or moral situations because they expect them to perform badly in these decisions (for more on task context see also Section 2.3).

Third, performance expectations, or the perception of the relative capabilities of the human and the automated agent may trigger algorithm aversion or appreciation irrespective of the context - and irrespective of whether this perception is accurate or not. An illustration of this comes from Logg et al. (2019), who replicate the airline passenger forecasting task first used in Dietvorst et al. (2015). The replication shows that when participants choose between themselves and the algorithm, they prefer to do the task themselves (as was the case in Dietvorst et al., 2015). In contrast, when participants choose between another human and the algorithm, a substantial part of them exhibits

algorithm appreciation (Logg et al., 2019). Hence, Logg et al. (2019) argue, (over-)confidence in one's own abilities may lead participants to exhibit algorithm aversion, yet doubts in the abilities of others may lead them to exhibit algorithm appreciation. This view is corroborated by a second difference between the two studies: in Dietvorst et al. (2015) participants have seen the algorithm make mistakes, while participants in Logg et al. (2019) received no information on performance prior to making their decision. Finally, the importance of perceived capabilities seems mirrored in the finding that experts are more likely to discard automated advice than are non-experts see e.g. Logg et al., 2019, experiment 4; Tazelaar & Snijders, 2004. Some additional evidence for the importance of performance expectations on the willingness to accept automated decision aids comes from Efendić, Van de Calseyde, & Evans (2020), who in a series of experiments document that humans are more willing to use human advice if it is given slowly, yet that the opposite applies to algorithmic advice: here, the participants were less likely to follow the advice if response time is higher. Interestingly to the present discussion, in one of these studies the authors, while focusing exclusively on automated advice, document that people would be more likely to disregard the automated advice and consult with a human colleague if response time is higher (study 6 in Efendić et al., 2020). Finally, while there is no comparison to accepting human advice, Filiz, Judek, Lorenz, & Spiwoks (2021) show that when participants receive feedback and are hence able to learn that the automated agent gives high-quality advice, the willingness to use its advice increases.

What emerges is that there is a clear need for more integrative studies to investigate which factors trigger the appreciation of, or aversion to, automated agents involved in decision-making situations. Without such studies, researchers run the risk of focusing only on one kind of behavior, and extrapolating it to too many situations, as is nicely illustrated by a survey contained in Logg et al. (2019): When the authors asked researchers in the field to predict how people would behave in their experiment, most researchers predicted that the participants would display algorithm aversion - yet the opposite happened. We hope that the hypotheses we have listed here could provide a good starting point for such research on the factors which influence the attitude of humans towards automated agents involved in decision-making.

5. Discussion and directions for future research

The behavioral research reviewed in this article provides a number of important insights into how humans interact with automated agents. However, many open questions remain, as Sections 3.3 and 4.4 have pointed out in detail. The importance of finding answers to these questions goes beyond a purely academic pursuit of knowledge. From an organizational standpoint, it is vital for firms to know how humans might interact with automated agents in various contexts. From a political standpoint, the recent technological advances have caused extensive discussions of the risks and benefits of automation (see e.g. Pasquale, 2015; Cowgill & Tucker, 2019). These discussions have been accompanied by repeated calls for regulation (see e.g. Citron & Pasquale, 2014; Doshi-Velez et al., 2017; Wachter & Mittelstadt, 2019). Both public opinion (Grzymek & Puntschuh, 2019) and policy-makers seem to support stricter regulation of automation, in particular with regards to AI see e.g. Wallace, 2020, for the EU, Roberts et al., 2019, for China, and OECD, 2019, for the OECD. The empirical research reviewed in this article may provide a number of insights for the development of such regulation, and for the development of optimal firm practices in an age of automation - as the following paragraphs will show with a number of examples.

Areas of application First, while the ever-moving technological frontier constantly offers new areas for automation, the research reviewed here shows that the question of where automation is beneficial cannot be answered solely on the basis of technical considerations. Indeed, the evidence shows that interacting with automated agents reduces the emotional and social response of humans. Hence, where social

rules or emotions form obstacles, automation may bring benefits beyond those of time and labour savings - for example, by increasing the willingness of people to disclose sensitive information like intimate partner violence (Ahmad et al., 2009; Humphreys et al., 2011). In contrast, in other situations social rules and expectations are desirable, and automation may prove detrimental. For example, studies on charitable giving have repeatedly shown that social image concerns and social pressure are important drivers of charitable giving (DellaVigna, List, & Malmendier, 2012; Andreoni, Rao, & Trachtman, 2017) - and one may hence speculate that automated solicitors might gather fewer funds than human ones. In an organisational context, due to the high importance of peer effects for worker performance (Mas & Moretti, 2009), replacing human peers with automated ones might decrease team performance.

Degree of automation A second insight concerns the optimal degree of automation. Numerous studies have documented that fully automated decisions can be more accurate and less discriminatory than human decisions (see e.g. Saez de Tejada Cuenca, 2019; Kleinberg, Lakkaraju, Leskovec, Ludwig, & Mullainathan, 2018; Stevenson, 2018; Hoffman, Kahn, & Li, 2018; Gates, Perry, & Zorn, 2002). Hence, one might argue, it would be beneficial to take humans out of the decision loop, thereby simplifying the decision problem to the task of creating the most accurate and least biased algorithm (see e.g. Miller, 2018). However, the studies reviewed here show that humans are particularly likely to exhibit algorithm aversion when they are - or when they feel that they are - replaced by automated agents. Entirely removing humans from the decision loop could thus make them particularly likely to mistrust and make inefficiently little use of algorithms. This empirical finding seems to be reflected in official government positions (see e.g. Hancock, 2015) and even in regulation (see e.g. art. 22(1) EU GDPR). However, further studies reviewed in this article also point to a potential downside of retaining human involvement in automated decision-making: people may over-rely on the recommendations produced by automated support systems, and fail to correct for the system's mistakes. For the moment, the reviewed evidence shows that the degree of human involvement constitutes an important factor which affects people's reaction towards automation. Further studies investigating this phenomenon will be necessary to solve the problem of the optimal degree of automation.

Accountability A particularly promising avenue of research in this regard seems to be the investigation of the impact of (real or perceived) accountability. Indeed, studies suggest that highlighting the responsibility of a human agent for the outcome of a decision could both decrease algorithm aversion by way of ensuring human intervention (see e.g. Dietvorst et al., 2018) and decrease automation bias by ensuring more effective monitoring (see e.g. Goddard et al., 2012). However, ensuring human accountability has recently become more difficult with the growing use of opaque, 'black-box' algorithms (see e.g. Pasquale, 2015; Lipton, 2018).

Transparency The obvious solution to this problem seems to lie in increased algorithmic transparency. Behavioral research can again provide valuable insights here. For instance, Erlei et al. (2020) shows that while transparency seems to contribute to the willingness to use a system in general, users seem to be less likely to incorporate the suggestions of a more transparent system into their actions (Erlei et al., 2020). Moreover, experiments show that transparency can also backfire; too much transparency can discourage people from using technology (Kizilcec, 2016; Lee, Jain, Cha, Ojha, & Kusbit, 2019), or create incentives to game the system (Ederer, Holden, & Meyer, 2018). In addition, transparency can be difficult to establish and have unforeseen effects. To illustrate this point, consider a particularly interesting finding from a computer science user study which develops an explanation method for a machine learning algorithm used to distinguish pictures of wolves and huskies (Ribeiro, Singh, & Guestrin, 2016). To test the explanations, the authors first showed a group of students trained in machine learning a number of classifications the algorithm had made. When the students were shown the raw data (the pictures) as well as the labels the algorithm had proposed, most of them stated that they trusted

the algorithm to perform well, even though they could see that it had made some mistakes. Only if, in addition, the authors showed the students an explanation of how the algorithm made these classifications (namely, that the snow in the background of the picture was a relevant factor) did the students lose trust in the algorithm - even though they had already known its accuracy rate before receiving the explanation. Hence, explanations matter: simply providing information about the accuracy of algorithm is not sufficient to enable observers to accurately judge an algorithm's validity. Indeed, a recent experiment shows that the way an algorithm is explained seems to affect how people perceive it (Cowgill, Dell'Acqua, & Matz, 2020).

More generally, transparency about automated decision-making can be misleading if there is no appropriate comparison to human decision-making. An example of this comes from a study which compares a CV screening algorithm and human HR managers (Cowgill, 2018). First, the author establishes that the algorithm gives a negative weight to candidates from non-elite universities. Thus, one might conclude, the algorithm is biased and harmful. However, it turned out that the candidates from non-elite schools actually disproportionately benefited from being screened by the algorithm, as the human evaluators would have assessed their credentials even more negatively. A similar finding comes from another study which finds that an automated recidivism risk score is biased, but not that the involvement of human decision-makers would result in any less biased decisions (Tan, Adebayo, Inkpen, & Kamar, 2018).

Perceptions of algorithmic decisions On top of the facts of how unbiased decisions are, perceptions of those decisions by people are important. People may perceive algorithmic decisions as less than fair even when they objectively are (Lee & Baykal, 2017) or think that automated decisions are more fair than identical human decisions (Bai et al., 2020). Further research will be needed to fully establish the impacts of transparency on automated decision-making.

6. Conclusion

The experimental evidence reviewed in this article shows that human-AA interactions are different from human-human interactions. In general, these interactions trigger less emotions, and are less affected - but not unaffected - by social concerns. Increasing the perception of the humanlikeness of automated agents through visual and behavioral cues tends to increase the level of emotional and social response they trigger. People seem more willing to interact with automated agents in contexts they perceive as analytical or objective - this seems to apply both to their attitude in interacting with automated agents and to their willingness to use automated decision-making support. In the workplace, humans have started interacting with automated agents who take the roles of teammates and managers. While the existing evidence clearly shows that the introduction of automated agents into the roles of teammates and managers has an effect on the behavior of humans in the workplace via a number of different channels, more careful consideration of when, for what tasks and with what type of human workers automated agents interact seems necessary before behavioral regularities and ensuing policy advice can be formulated. Another area in which many open questions remain is automated decision-making: while some experiments find that people prefer and (over-)rely on automated advice, rather than human advice, others find that they are averse to automated decision-makers replacing human ones. We have discussed a number of factors which might drive these results, including the distribution of agency in the decision-making situation, the context of the decisions and performance expectations. In summary, the reviewed studies not only show what we know about human-AA interactions, but also point out a number of open questions and avenues for future research.

Our hope is that this review and discussion can serve to inform decision-makers and managers who consider using automated agents in their practices. With the evidence gathered here, and by highlighting potential inconsistencies and avenues for future research, we aim to

further stimulate the lively research on this topic across disciplinary boundaries.

Declaration of Competing Interest

None.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.jsocec.2022.101897](https://doi.org/10.1016/j.jsocec.2022.101897).

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